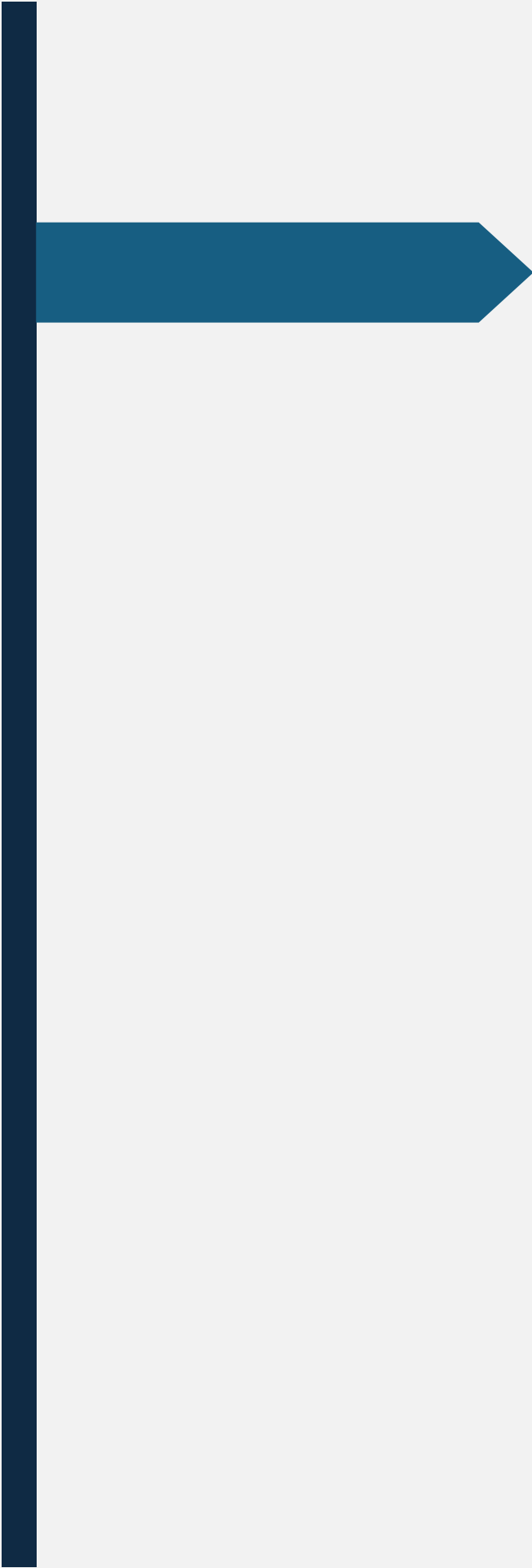




University Campus

Network Design & Implementation

3-Tier Topology | Cisco Packet Tracer

Under Supervision:

Eng. Khaled Eid

1. Project Overview

This project is a complete campus network for a university, designed and simulated in Cisco Packet Tracer. It follows a redundant 3-tier model (core, distribution, access) and serves four departments, each in its own VLAN and subnet: Management, Staff, Students, and Servers. Redundant links, link aggregation, and first-hop redundancy keep the network available if a device or link fails, while dynamic routing, ACLs, and NAT handle connectivity and security.

Topology at a glance

- Router0 connects to two Core switches over the 10.0.50.0/24 and 10.0.60.0/24 links.
- Four multilayer distribution switches (two on the left, two on the right) are cross-connected to both cores.
- Six access switches connect the Management, Staff, and Students end devices.
- A dedicated server switch hosts the DHCP, WEB, and DNS servers in VLAN 40.
- Redundant parallel links between the core switches and between each distribution switch pair are bundled with EtherChannel.

2. Technologies Implemented

Technology	Purpose in this project
VLANs	Segment the campus into four broadcast domains: Management (10), Staff (20), Students (30), and Servers (40).
Inter-VLAN Routing	Layer 3 switches route traffic between VLANs so every department can reach the others and the servers.
HSRP	Provides a virtual default gateway per VLAN (10.0.X.1), so hosts keep their gateway if one distribution switch fails.
EtherChannel	Bundles parallel physical links into one logical link for more bandwidth and link-level redundancy.
STP	Prevents Layer 2 loops in the redundant, cross-connected topology while keeping backup paths ready.
OSPF	Link-state routing protocol that advertises all VLAN subnets and core links and converges quickly.
RIP	Distance-vector routing protocol (hop-count metric) configured as an additional routing method.
ACLs	Filter traffic between VLANs and toward the servers to enforce the campus security policy.
NAT	Translates the private 10.0.0.0/8 addressing to a public address at the edge router for outside access.
DHCP	The DHCP server in VLAN 40 automatically assigns IP settings to hosts in all VLANs (via relay).
DNS	The DNS server resolves hostnames so users reach services by name instead of IP address.
Web Server	Hosts the campus website, reachable from every VLAN.

3. Subnetting Table and Default Gateways

Department	Network Address	Subnet Mask	Hosts	Default Gateway
Management	10.0.10.0	255.255.255.0	254	10.0.10.1
Staff	10.0.20.0	255.255.255.0	254	10.0.20.1
Students	10.0.30.0	255.255.255.0	254	10.0.30.1
Servers	10.0.40.0	255.255.255.0	254	10.0.40.1
Router Gig0/0 link	10.0.50.0	255.255.255.0	254	-
Router Gig0/1 link	10.0.60.0	255.255.255.0	254	-

4. VLAN Table

VLAN Name	ID	Subnet	Default GW (HSRP VIP)	Interfaces
Management	10	10.0.10.0/24	10.0.10.1	Fa0/3-4
Staff	20	10.0.20.0/24	10.0.20.1	Fa0/5-7
Students	30	10.0.30.0/24	10.0.30.1	Fa0/8-13
Servers (DHCP, WEB, DNS)	40	10.0.40.0/24	10.0.40.1	Fa0/2-4

5. Routing and Redundancy

OSPF

OSPF is a link-state dynamic routing protocol that calculates the best path between all subnets (VLANs). It runs on the Layer 3 devices (router and core/multilayer switches), which advertise every VLAN subnet and inter-device link to build a complete network map. The Shortest Path First (SPF) algorithm selects the optimal, loop-free route using link cost (bandwidth), and Link-State Advertisements (LSAs) give fast convergence when the topology changes. The router advertises the core links 10.0.50.0 (Gig0/0 to the left core switch, Fa0/9) and 10.0.60.0 (Gig0/1 to the right core switch, Fa0/9), which provides redundancy and efficient inter-VLAN routing across the 3-tier topology.

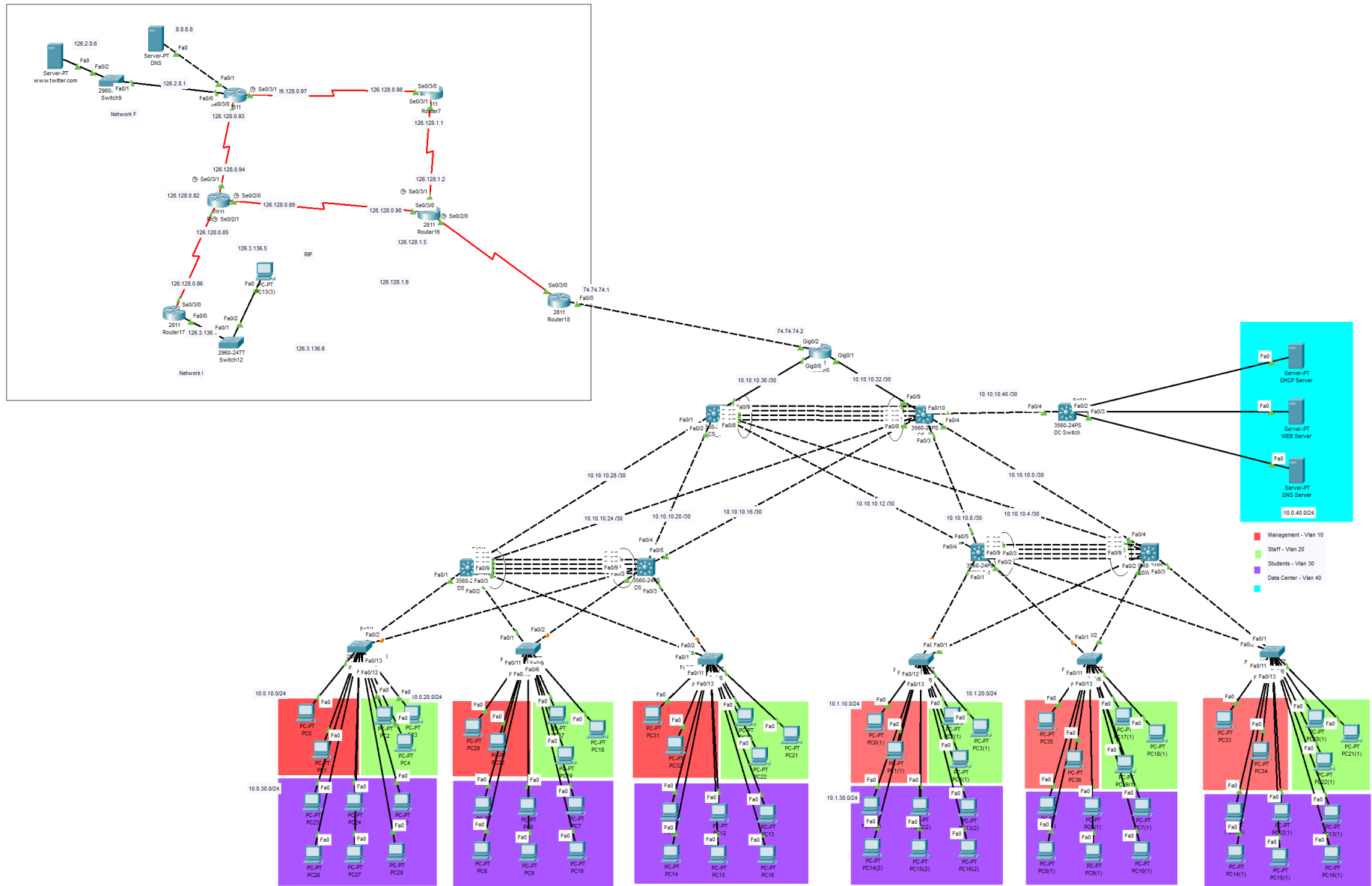
RIP

RIP is a distance-vector protocol that uses hop count as its metric (maximum 15 hops) and shares its routing table with neighbors periodically. It is simple to configure, but converges more slowly and scales less well than OSPF, which is why OSPF is the primary protocol for the campus core.

HSRP, EtherChannel, and STP

HSRP gives each VLAN a virtual gateway IP (10.0.10.1, 10.0.20.1, 10.0.30.1, 10.0.40.1) shared by a pair of distribution switches, so one active switch forwards traffic and the standby takes over automatically on failure. EtherChannel combines the parallel links between switch pairs into a single logical link, and STP blocks redundant Layer 2 paths to prevent loops while keeping them available as backups.

6. Logical Diagram



7. Packet Tracer File: download the fully configured topology from [here](#).